

Restaurant Review Analysis and Recommendation

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Course Project

GitHub: <https://github.com/afnanwioe456/DS-Final-Project.git>

Abstract—This project asks whether large collections of restaurant reviews can be turned into interpretable evidence for recommendation and feedback analysis without requiring manual labeling. The central objective is not only to predict ratings, but also to explain why a review receives a particular score and what experience factors matter most for analysis and operational monitoring. To address this problem, the project builds a data science pipeline that combines data filtering, text normalization, sentiment scoring, latent topic discovery, and aspect-level explanation. The corpus contains 17,318 restaurants and 72,125 merged reviews from the Yelp Academic Dataset. Empirically, review sentiment is moderately and positively correlated with star ratings ($r = 0.608$), while review length is weakly negative. Topic modeling exposes coherent experience factors such as food quality, service quality, waiting time, value, and mixed dining experiences. A linear regression model using topic probabilities, sentiment, review length, and business-level stars reaches a test R^2 of 0.462, and a weighted variant reduces test R^2 to 0.352. A transformer-based aspect module provides finer-grained explanations for food, service, ambiance, and price on sampled reviews. Overall, the results show that interpretable review analytics can capture both aggregate satisfaction and specific complaints, which is useful for recommendation and operational feedback.

Index Terms—data science, restaurant reviews, topic modeling, sentiment analysis, aspect-based explanation, recommendation, explainable NLP

I. INTRODUCTION

Online restaurant reviews contain a large amount of useful but noisy behavioral evidence. A single review may describe food quality, service, waiting time, atmosphere, price, and the reviewer’s overall emotion in one short paragraph. This text can support recommendation, quality monitoring, and descriptive analysis that are not visible in aggregate star ratings alone. The project therefore studies how unstructured review text can be converted into structured signals for rating explanation and feedback analysis.

The motivating assumption is that an interpretable system should do more than output a score. It should preserve the reasons behind the score in an inspectable form and support downstream analysis. In this setting, the task is to build a review analysis pipeline that connects raw text to latent experience factors and then to rating explanation. The project goal is to identify recurring themes in reviews, quantify their relationship with star ratings, and produce explanations that correspond to common restaurant attributes rather than abstract model parameters.

This problem matters for several reasons. First, review platforms are large enough that manual reading is impractical. Second, star ratings alone are too compressed to capture mixed experience. A review can be generally positive while still complaining about delays or poor service. Third, aggregate prediction alone does not explain the variability in user feedback. The project addresses these issues by treating review text as a source of experience evidence rather than as a classification target in isolation.

The main findings are consistent across several analysis stages. Sentiment is strongly aligned with ratings, topic discovery recovers interpretable experience clusters, and aspect-level analysis isolates food, service, ambiance, and price as the main explanatory dimensions. A simple linear model already explains a substantial portion of rating variance, while more detailed aspect-based explanations improve interpretability on individual examples. The work therefore shows that data science can extract structured signals from noisy review corpora.

II. RELATED WORK AND BACKGROUND

The project builds on standard methods for organizing unstructured language into interpretable representations. Topic models such as Latent Dirichlet Allocation (LDA) represent each document as a mixture of latent topics and each topic as a distribution over words [1]. This is a natural fit for review corpora because one review often contains multiple experience dimensions. TF-IDF weighting is used to suppress generic terms that appear in many documents but contribute little semantic discrimination [2]. The project also uses lexicon-based sentiment scoring and a pretrained transformer for zero-shot aspect classification. The transformer model is based on BART-MNLI, which is a general-purpose pretrained language model that can be reused for classification tasks without task-specific labeling [3].

From an interpretability perspective, the key idea is that model outputs should remain legible to non-technical users. A topic label such as *poor service and negative experience* is more useful than an opaque latent dimension index. Likewise, a review explanation that says food quality contributed positively while service contributed negatively is easier to act on than a raw regression coefficient. This framing aligns the technical pipeline with decision support, not only prediction.

III. DATA

The data come from the Yelp Academic Dataset and combine the business and review tables. The analysis filters the business table to restaurant-related entries and then merges those businesses with their reviews. After filtering, the working dataset contains 17,318 restaurants and 72,125 reviews. This scale is large enough to support both exploratory analysis and downstream modeling, while still allowing topic inspection and manual interpretation.

The corpus has two notable properties. First, the restaurant review distribution is heavily skewed toward high ratings, which is typical for public review platforms. Second, review length is long-tailed: most reviews are short, while a smaller number provide detailed explanations or complaints. These properties matter because they affect both model training and evaluation. A method that performs well on average may still fail on long, detailed, or negative reviews, which are often the most informative for analysis.

Preprocessing is intentionally conservative and reproducible. The baseline cleaning step lowercases text and removes punctuation and numbers. A more refined variant adds lemmatization and a custom stop-word list to suppress high-frequency review vocabulary that blurs topic boundaries. This step is important because restaurant reviews contain many generic positive words such as *good*, *great*, and *place*, which can dominate topic formation without adding interpretive value. For modeling, the project compares count-based and TF-IDF vectorizations. The final regression features combine topic probabilities, sentiment polarity, review length, and business-level stars.

Figure 1 summarizes the main descriptive properties of the corpus. The rating distribution is concentrated in the upper range, the review-length distribution is long-tailed, and sentiment shows a clear positive association with ratings.

IV. GEOGRAPHIC COVERAGE

Although the main analysis is review-centric rather than city-centric, the dataset is still geographically diverse. The most frequent cities in the merged corpus include large and mid-sized restaurant markets, which indicates that the analysis is not restricted to one local scene. This matters because review language, restaurant type, and expected service quality can differ across locations. In a broader system, city-level effects would be a useful extension because they can capture regional cuisine patterns, competition density, and different review norms.

Figure 2 shows the distribution of the ten most common cities in the working dataset. The concentration in several major metro areas suggests that the corpus is large enough for cross-city analysis, but the current project deliberately keeps the modeling at the review level to preserve interpretability. A location-aware extension could introduce city as an explicit control variable, or it could train separate models for major metropolitan clusters.

V. METHOD

The full pipeline is organized as a sequence of interpretable transformations rather than a single monolithic model. Figure 3 provides an overview.

A. Preprocessing and feature construction

The preprocessing stage standardizes review text and prepares it for both unsupervised and predictive modeling. The baseline representation uses cleaned unigrams in a bag-of-words space. The refined representation uses TF-IDF with thresholds that filter out very common and very rare words. This is important for review corpora because a large fraction of tokens are either ubiquitous sentiment words or domain-specific noise. The result is a more compact vocabulary that emphasizes semantically meaningful terms.

B. Sentiment scoring

Each review is assigned a continuous sentiment polarity score. This score is not treated as a final answer, but as a compact summary of the overall tone of the review. In the downstream analysis it serves three purposes: it provides a simple correlation baseline, it supplies one feature in the rating model, and it helps separate positive from negative reviews inside topic clusters.

C. Topic discovery

LDA is used to discover latent experience factors in the review text. The model treats each review as a mixture over latent topics and each topic as a distribution over words. In the final configuration, the topic model is fit on the refined TF-IDF representation, and the resulting topics are manually labeled according to the most salient words. The goal is not to produce a psychologically perfect latent structure, but to recover categories that correspond to common restaurant experiences: food quality, service, value, atmosphere, and negative service episodes.

A second, stricter configuration is used as a robustness check. By tightening the document-frequency thresholds, the model reduces lexical noise and reveals whether the discovered structure remains stable when the vocabulary is simplified. This is useful because review corpora often contain generic adjectives that can overwhelm semantically distinct terms.

D. Rating prediction

To connect textual evidence to star ratings, the project trains a linear regression model on standardized features:

$$\hat{y} = \beta_0 + \sum_{k=1}^K \beta_k z_k,$$

where the feature vector contains topic probabilities, sentiment score, review length, and business-level stars. The linear form is intentionally simple. It is easy to inspect,

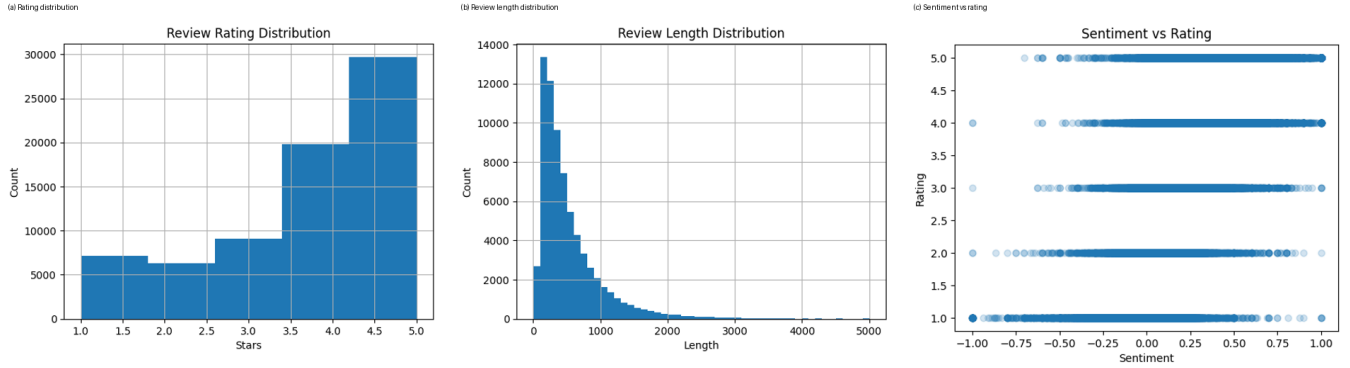


Fig. 1. Core data diagnostics. The corpus is positively skewed in ratings, review length is long-tailed, and sentiment tracks rating directionally.

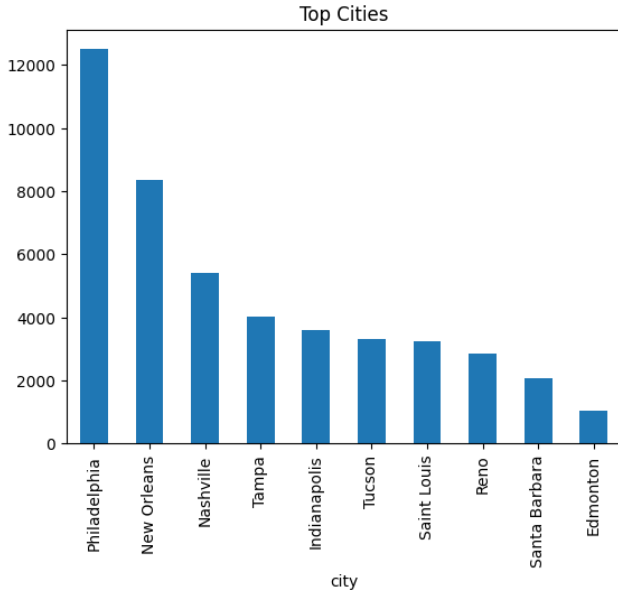


Fig. 2. Top cities in the merged review corpus. The data cover several major restaurant markets, which makes location-aware modeling a plausible next step.

easy to reproduce, and easier to explain than a black-box model. A weighted variant is also trained using class-balanced sample weights derived from rounded ratings, primarily to test whether performance improves when the target distribution is rebalanced.

E. Aspect-level explanation

The project also uses a pretrained zero-shot classifier to estimate aspect-level sentiment for a small review sample. The aspects are food quality, service, ambiance, and price. For each aspect, the model predicts a polarity label and confidence score. A directional score is then formed by multiplying the polarity sign by its confidence:

$$s_a = \text{sign}(p_a) \cdot c_a.$$

These aspect scores are used in two ways. First, they support a more fine-grained linear model for rating ex-

planation. Second, they are analyzed with PCA to test whether the aspect space has a compact low-dimensional structure.

VI. EXPERIMENTS AND EVALUATION

The evaluation strategy combines descriptive analysis, unsupervised inspection, predictive modeling, and qualitative explanation checks. The project does not frame the task as a single classification benchmark. Instead, it asks whether the extracted representations are both predictive and interpretable.

The first baseline is the direct sentiment–rating relationship. Pearson correlation is appropriate because both variables are continuous and the goal is to assess directional agreement. The second baseline is the linear regression model trained on topic and metadata features. Here, R^2 measures how much of the rating variance is explained by the model, while MAE provides an error measure in rating units. For the unsupervised topic stage, perplexity is used as a coarse stability signal, recognizing that topic quality also needs manual review. For the aspect module, PCA explained variance is used to check whether the four aspect dimensions are highly redundant or genuinely multi-dimensional.

This metric set is appropriate for an interpretable setting. A purely predictive score would ignore explanation quality, while a purely qualitative analysis would not show whether the representations carry enough signal to support downstream use. The chosen metrics therefore balance utility, structure, and interpretability.

VII. RESULTS

Table I shows the core statistical relationship between text-derived signals and ratings. Sentiment is positively correlated with rating, and longer reviews tend to be slightly more negative or mixed. The sign pattern is consistent with common review behavior: short reviews are often simple endorsements, while longer reviews more frequently contain balanced praise and criticism.

Topic discovery produces coherent and business-relevant clusters. Figure 4 combines the average rating by topic

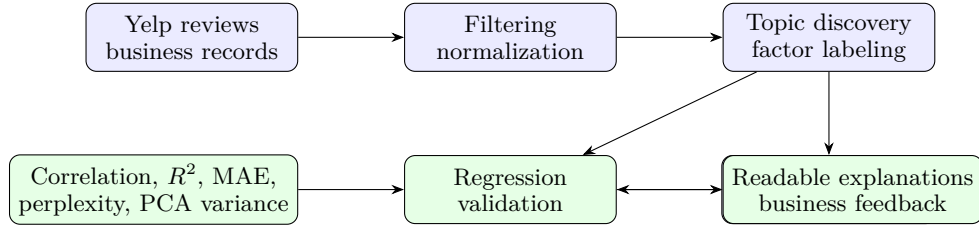


Fig. 3. Pipeline overview. Each stage preserves an interpretable representation of the review evidence.

TABLE I
CORRELATION AMONG THE MAIN REVIEW FEATURES.

	Sentiment	Rating	Review Length
Sentiment	1.000	0.608	-0.228
Rating	0.608	1.000	-0.156
Review Length	-0.228	-0.156	1.000

with the corresponding word-cloud summary. The highest-rated topic emphasizes food quality and positive dining language, while the lowest-rated topic centers on waiting, ordering, and service problems. This is a useful result because it shows that topic discovery is not just separating vocabulary statistically; it is recovering recognizable experience patterns.

Table II lists the final topic labels, representative words, mean ratings, and counts. The corpus is dominated by a broad service-and-casual-dining cluster, while the rarest clusters correspond to value and drinks/social dining. The imbalance is informative: it suggests that ordinary service narratives are extremely common, but they are not sufficient to explain rating variation on their own.

The primary regression model reaches a train R^2 of 0.456 and a test R^2 of 0.462. This is a meaningful result given the simplicity of the model and the noisy nature of review text. The weighted variant does not improve generalization; its test R^2 falls to 0.352. This suggests that rebalancing the target distribution makes the model less fit to the natural rating frequencies, even if it changes the coefficient profile.

Figure 5 shows the standardized regression coefficients and the aspect sentiment distributions. Sentiment is the strongest positive coefficient, followed by business-level stars. Topic 2 also contributes positively, while Topic 4 is clearly negative, matching the service-complaint interpretation. On the aspect side, food quality and price show the largest spread, while service and ambiance are somewhat more concentrated around positive values. This indicates that the aspect classifier is capturing useful differences in review tone.

Table IV reports test performance by user activity. The model fits low-activity users best and performs worse on highly active users. This pattern is plausible because frequent reviewers tend to have more idiosyncratic standards and more varied writing styles. The finding matters from

a interpretability perspective because a single aggregate model may not serve all user segments equally well.

The aspect-based PCA plot in Figure 6 shows that the four directional aspect scores occupy a highly structured low-dimensional space. The first two principal components explain 97.0% of the variance, which indicates that the aspect representation is compact and largely redundant in its original form. This is useful because it means a small set of interpretable dimensions already carries most of the variation in the sample.

Table V provides two qualitative explanations. The first example is a clearly positive review: food quality is the strongest positive contributor, and the predicted rating slightly exceeds the observed rating. The second example is mixed to negative: food quality, price, and service all contribute negatively, which lowers the predicted score. These examples show how the aspect module turns a score into a readable rationale.

VIII. ROBUSTNESS AND ERROR ANALYSIS

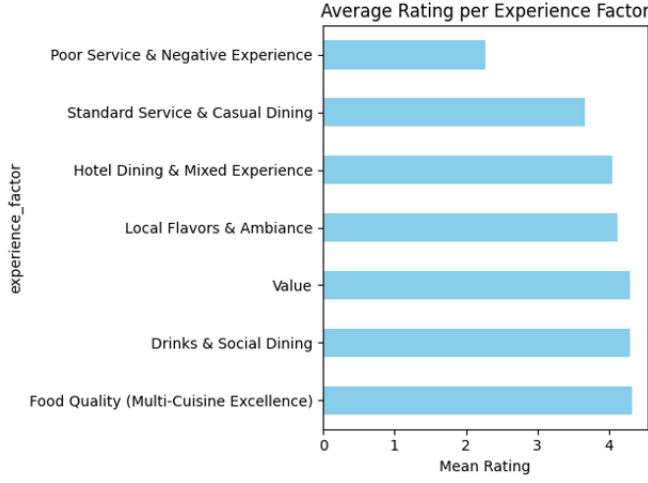
The project included several robustness checks because review corpora are highly sensitive to preprocessing choices. A small change in vocabulary filtering can shift the topic structure substantially, and a small change in sample weighting can change the regression objective. These checks are important in an interpretable setting because the final artifact is meant to support explanation, not merely optimize a single score.

A. Topic configuration sensitivity

The initial topic model was trained on a simple count-based representation with five topics. It already produced coherent clusters, but several topics mixed food vocabulary with generic service words. The refined TF-IDF configuration improved interpretability by suppressing very common terms and allowing more distinctive terms to appear near the top of each topic. The final seven-topic model is therefore the best balance between granularity and readability in this project. A stricter TF-IDF configuration was also tested as a stress check. Under that setting, the topic mass collapsed into two dominant clusters and several very small clusters, which indicates that the corpus has a strong shared narrative but only a limited number of rare submodes.

Table VI summarizes how the main experimental choices affected the analysis. The table is not intended as a formal

(a) Mean rating by experience factor



(b) LDA topic word clouds



Fig. 4. Topic discovery summary. The topic labels align with recognizable restaurant experience factors, and the mean ratings differ meaningfully across factors.

TABLE II
INTERPRETED TOPIC FACTORS AND AVERAGE RATINGS.

Topic label	Representative words	Mean rating	Count
Food Quality (Multi-Cuisine Excellence)	chicken, delicious, sushi, best, taco, fresh, love, service, roll, amazing	4.325	16,321
Drinks & Social Dining	sangria, tour, flatbread, empanadas, quiche, chaat, cider, los, habit, grandma	4.296	54
Value	prime, downside, oyster, doughnut, coke, basket, cheesesteaks, cent, boiled, dined	4.286	7
Local Flavors & Ambiance	beignet, crepe, dan, website, lighting, indianapolis, stl, desired, impeccable, funky	4.118	17
Hotel Dining & Mixed Experience	hotel, gyro, room, relaxed, cheesecake, greek, cupcake, annoying, coworker, maple	4.042	24
Standard Service & Casual Dining	service, pizza, order, nice, burger, drink, staff, table, ordered, bar	3.660	55,658
Poor Service & Negative Experience	min, poor, hibachi, milkshake, sushi, rude, tonight, zero, anniversary, manager	2.273	44

TABLE III
LINEAR REGRESSION RESULTS FOR RATING PREDICTION.

Model	Train R^2	Test R^2
Unweighted linear regression	0.456	0.462
Weighted linear regression	0.347	0.352

TABLE IV
TEST PERFORMANCE BY USER ACTIVITY GROUP.

Activity group	n	MAE	R^2
Low (1–2 reviews)	12,426	0.789	0.480
Medium (3–5 reviews)	1,455	0.747	0.290
High (6+ reviews)	544	0.682	0.251

TABLE V
QUALITATIVE EXPLANATION EXAMPLES FROM THE ASPECT-BASED MODULE.

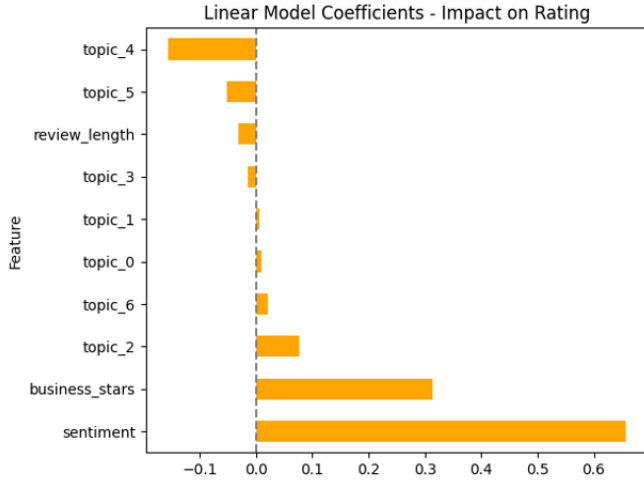
Case	Actual / Predicted	Dominant aspects	Interpretation
A	4.0 / 4.51	Food, price, ambiance, service positive	Strongly favorable review with broad positive tone
B	3.0 / 3.25	Food, price, service negative; ambiance weakly positive	Mixed review with dissatisfaction focused on core dining experience

hyperparameter sweep. Rather, it records the behavior of the most important design decisions so that the analysis can be reproduced and interpreted correctly.

B. Why weighted regression underperformed

The weighted regression experiment reduced test R^2 from 0.462 to 0.352. This is not surprising: balanced

(a) Standardized regression coefficients



(b) Aspect sentiment distributions

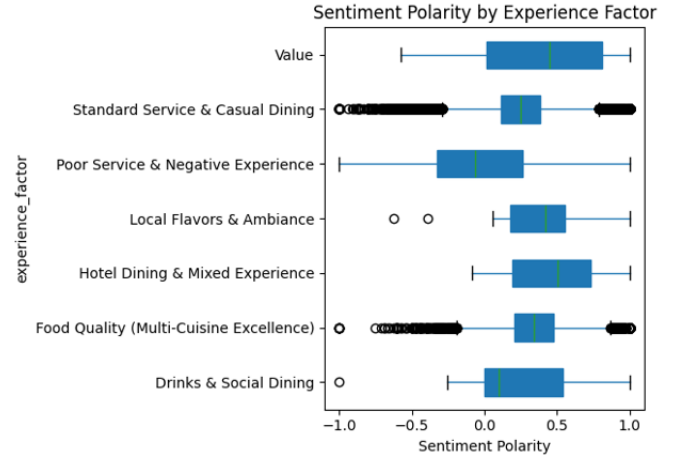


Fig. 5. Predictive and explanation diagnostics. Sentiment and business stars dominate the linear model, while aspect-level scores separate food, service, ambiance, and price effects.

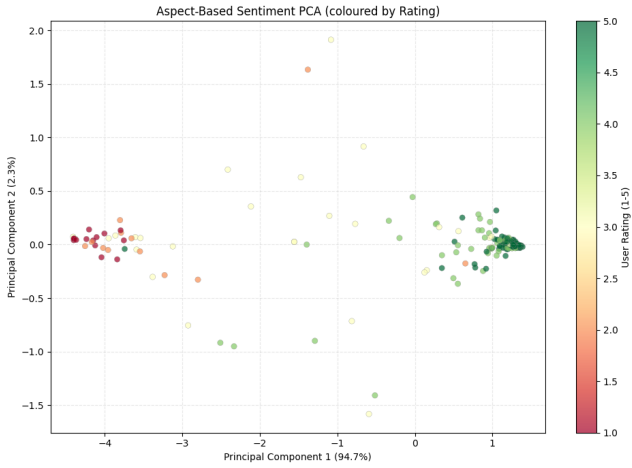


Fig. 6. Aspect-based sentiment PCA. The first two components explain 97.0% of the variance, indicating strong low-dimensional structure.

weighting re-optimizes the model toward rare rating levels, which increases pressure to fit cases that are underrepresented in the natural distribution. In effect, the model gives up some overall fit to reduce bias toward frequent positive ratings. From a decision-support perspective this trade-off may still be defensible if the goal is to highlight dissatisfied users, but it is not advantageous when the goal is overall rating prediction.

C. Qualitative failure cases

The explanation module is useful, but it is not perfect. A positive food signal can outweigh a weaker complaint about service, producing a forecast that looks overly optimistic for mixed reviews. Conversely, a service-heavy review can be forced into the dominant service cluster even when the central issue is price or waiting time. These error

TABLE VI
ROBUSTNESS CHECKS ACROSS REPRESENTATIONS AND MODELS.

Configuration	Representation	Observed behavior	Use
Count-based LDA (5 topics)	Bag-of-words	Coarse but readable topics; several clusters still mixed generic and domain-specific terms	Exploratory baseline
TF-IDF LDA (7 topics)	Refined vocabulary	Best balance between granularity and interpretability; clear topic-rating separation	Final topic analysis
TF-IDF LDA (strict 5 topics)	Strong filtering	Two dominant clusters absorbed most reviews; rare topics became nearly empty	Stability check
Unweighted regression	Topic probabilities + sentiment + metadata	Best test fit; coefficients easy to inspect	Main predictive baseline
Weighted regression	Balanced sample weights	Lower generalization accuracy; stronger emphasis on rare ratings	Sensitivity test

modes are expected because both the topic model and the aspect classifier compress a rich review into a small number of dimensions. The main implication is that the outputs should be used as interpretable summaries rather than as ground truth.

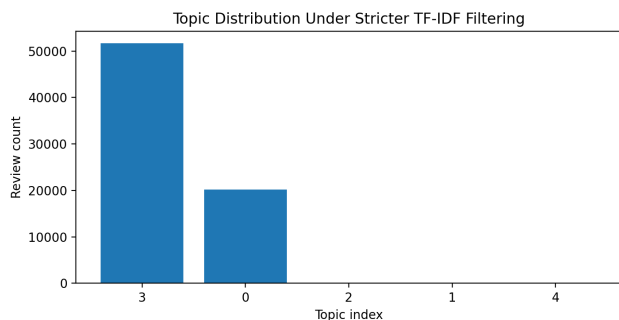


Fig. 7. Topic distribution under stricter TF-IDF filtering. The corpus concentrates heavily in the two largest clusters, which shows that rare topics are fragile under stronger vocabulary constraints.

D. Implications for deployment

For a practical review dashboard, the most reliable use of the current pipeline would be aggregation rather than one-shot decision making. Topic-level summaries can identify recurring complaints across many reviews, sentiment can provide a quick trend indicator, and aspect-level scores can be attached to individual comments for manual inspection. The model is less suitable as an autonomous ranking engine for restaurants because it does not model user taste diversity, neighborhood effects, or restaurant category differences. The current analysis therefore serves best as an explanation layer on top of recommendation or monitoring infrastructure.

IX. DISCUSSION

The project succeeded in three places. First, it converted a noisy review corpus into a small set of interpretable experience factors. Second, it established that these factors are connected to ratings in a measurable way. Third, it produced explanations that are readable at the level of restaurant experiences rather than machine features. From an interpretability perspective, this is the key value of the pipeline: the analyst can inspect not only *what* the model predicts, but also *why*.

The results also reveal several limitations. Topic modeling is sensitive to generic vocabulary, and some clusters still mix food, service, and broad positivity language. This is especially visible in the dominant service-and-casual-dining topic, which is too broad to be a fully clean semantic category. The weighted regression experiment also shows that trying to balance the rating distribution can reduce predictive fit. In other words, improving fairness across the target distribution may come at a cost in aggregate accuracy. The project therefore demonstrates a genuine trade-off rather than a one-directional improvement.

Another limitation is that the aspect-based module is computationally expensive and therefore applied only to a small sample. This means the qualitative explanations are useful but not yet scalable to the full corpus in their

current form. A production-quality version would need batching, caching, or a lighter classifier. The model is also not personalized: it does not adapt to individual reviewer style or restaurant type. The lower performance on high-activity users suggests that a single global model is too coarse for some segments.

Despite these issues, the analysis provides a solid interpretable foundation. The discovered topics are usable as business-facing categories, the regression model provides a transparent numeric summary, and the aspect module adds case-level interpretability. In a practical setting, this kind of pipeline could support review dashboards, complaint summaries, and explanation-aware recommendation systems.

X. IMPLEMENTATION AND REPRODUCIBILITY NOTES

The implementation was intentionally kept simple enough to reproduce with standard Python libraries. The corpus was processed in a notebook workflow using common data-science tools for parsing JSON records, cleaning text, vectorizing documents, and fitting the downstream models. The topic model and regression stages use fixed random seeds so that the main outputs are stable across runs. This is important because both LDA and sample-based explanation methods can otherwise produce slightly different outputs each time they are executed.

Two parameter settings were used repeatedly in the analysis. The exploratory model used a 5-topic LDA configuration on count vectors, while the refined analysis used a 7-topic LDA configuration on TF-IDF features. In the second stage, the document-frequency thresholds were tightened to test whether rare topics remained interpretable under stronger filtering. For the predictive model, the feature set was standardized before fitting the linear regression so that coefficient magnitudes could be compared directly. For the aspect module, the analysis was limited to a small sample because zero-shot transformer inference is much slower than the earlier stages.

These implementation choices are not arbitrary. They reflect a trade-off between scale, interpretability, and runtime. The early pipeline stages are fast enough to run on the full corpus, which makes them appropriate for exploratory dashboards. The later aspect stage is more expensive but yields richer local explanations, which makes it better suited to sampled case analysis or offline reporting. The final report therefore separates corpus-wide structure from review-level explanation rather than forcing both into the same model.

XI. CONCLUSION

This project studied whether restaurant reviews can be transformed into interpretable evidence for recommendation and feedback analysis. Using Yelp review and business data, the pipeline combined preprocessing, sentiment scoring, topic discovery, linear prediction, and aspect-level explanation. The most important findings are that sentiment

strongly tracks rating, latent topics recover recognizable dining experiences, and a simple regression model explains a substantial portion of rating variance. The aspect-based module further shows that food quality, service, ambiance, and price can be converted into readable explanations for individual reviews.

The work also clarifies what remains difficult. Review corpora are noisy, topic boundaries are imperfect, and frequent reviewers are harder to model with a single global form. With more time, the strongest next steps would be improved domain-specific preprocessing, a more stable topic representation, broader aspect coverage, and a formal evaluation of explanation usefulness. Even in its current form, however, the project demonstrates that data science can make large-scale review text more interpretable, more actionable, and more useful for decision support.

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